

A deterministic obstruction to a universal subadditive rate

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Conjecture 00000008492

Both source versions assert that the universal spectrum of rates of mean convergence for random subadditive sequences is dominated by $\log n/n$. We disprove this assertion even on a one-point ergodic probability space. In particular, the obstruction persists when the implied constant and the starting index may depend arbitrarily on the process.

Theorem 1. *There is an integrable, nonnegative subadditive process on an ergodic probability-preserving system whose normalized expectations converge to zero, but whose errors are not $O(\log n/n)$.*

Proof. Let $\Omega = \{*\}$, let μ be the unit point mass, and let T be the identity. This system preserves probability and is ergodic: its only subsets are the empty set and Ω . Define, for every integer $n \geq 0$,

$$X_n(*) = \sqrt{n}.$$

Every X_n is a measurable integrable random variable, is nonnegative, and $X_0 = 0$. For all $m, n \geq 0$,

$$X_{m+n}(*) = \sqrt{m+n} \leq \sqrt{m} + \sqrt{n} = X_m(*) + X_n(T^m *).$$

The inequality follows by squaring the nonnegative quantities: the square on the right is $m + n + 2\sqrt{mn} \geq m + n$. Thus this is the usual subadditive cocycle inequality, with its actual iterate of the measure-preserving map.

Since integration against μ evaluates at its one point,

$$a_n := \frac{\mathbb{E}_\mu X_n}{n} = \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} \longrightarrow 0 \quad (n \geq 1).$$

The normalized process has the same limit pointwise and in mean. Its one-sided error from the limiting value 0 is exactly $a_n > 0$.

For any real constant C , the standard logarithm–power limit gives

$$C \frac{\log n}{\sqrt{n}} \longrightarrow 0.$$

Consequently, for every starting index N , there exists $n \geq \max(N, 1)$ such that $C \log n < \sqrt{n}$. Dividing by $n > 0$ yields

$$C \frac{\log n}{n} < \frac{\sqrt{n}}{n} = a_n.$$

This excludes every eventual big- O constant, even for this single fixed process, and proves the theorem. $\square$

Scope of the disproof

The source concerns random subadditive sequences and does not impose a uniform bound on all unnormalized X_n , mixing assumptions, or a special symbolic construction. The example is deterministic, which is a valid special case of a random process, and its underlying probability system is ergodic. Each random variable is bounded and integrable; the normalized variables are uniformly bounded by 1 for $n \geq 1$. The unnormalized values grow as $\sqrt{n}$. No claim is made about a different hypothesis requiring $\sup_n \|X_n\|_\infty < \infty$.

The assertion about a universal upper rate is a necessary clause of the conjecture. Its failure already disproves the conjunction; no assertion about the separate Sturmian or Thue–Morse extremal constructions is needed. The counterexample does not merely violate a coefficient-one inequality: it violates every possible eventual constant multiple of the proposed rate.

Formal correspondence

The accompanying Lean project uses Mathlib’s actual probability, ergodicity, integration, filter convergence, and asymptotic definitions.

- `mu` (written μ in the source) is `Measure.dirac ()` on `Unit` with its full sigma-algebra. An `IsProbabilityMeasure` instance and `system_ergodic` establish the probability and `Ergodic T mu` conditions.
- `X n` is the actual real square root of the natural index. `integrable_X`, `X_nonneg`, `X_zero`, and `subadditive_process` verify integrability, nonnegativity, the zero initial value, and the full iterated subadditive inequality.
- `expectation_X` computes the Bochner integral. `mean` divides that integral by the index; `mean_tendsto_zero` and `pointwise_normalized_limit` prove convergence, using the real negative-power limit.
- The logarithm little- o power theorem yields the limit of $\log n/\sqrt{n}$. This is formalized as `log_div_sqrt_tendsto_zero`. The theorem `no_eventual_rate` proves

$$\forall C \in \mathbb{R} \forall N \in \mathbb{N} \exists n \geq N, \quad n \geq 1 \quad \text{and} \quad C \frac{\log n}{n} < a_n.$$

- `not_big0` proves the negation of Mathlib’s `Asymptotics.IsBig0` at `atTop`, including its norm-based definition. Positivity identifies the formal error with the one-sided error.

Natural-index division at $n = 0$ uses Lean’s totalized field convention; all counterexample witnesses have $n \geq 1$, so this has no asymptotic effect.

Reproduction

The project pins Lean 4.19.0 and Mathlib commit

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

From `lean/`, run `lake update`, `lake exe cache get`, and `lake build`. Check the complete source with

```
lake env lean -DwarningAsError=true Main.lean.
```

The source prints axiom audits for the principal theorems. Only standard logical axioms are used; there are no admitted statements or additional axioms.